

Hallucination Detection Report

Generated: 2026-04-16 00:12 · Extracted triples: 8 · Verdicts: 6 · Inconsistencies: 0 · Symbolic violations: 2

Executive Summary

2 Correct	2 Predicate Mismatch	2 Fabricated Fact
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Extracted Triples

All verifiable triples extracted from the LLM response before any canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
575	HAS_INTEREST_RATE	12.21%	Yes	FACTUAL_CLAIM	Indicates the interest rate associated with the subject.
580	HAS_INTEREST_RATE	11.91%	Yes	FACTUAL_CLAIM	Indicates the interest rate associated with the subject.
579	HAS_INTEREST_RATE	12.03%	Yes	FACTUAL_CLAIM	Indicates the interest rate associated with the subject.
579	HAS_RISK_TIER	deep subprime	Yes	FACTUAL_CLAIM	Specifies the risk tier classification related to the subject.
890	HAS_INTEREST_RATE	Below 4.36%	Yes	FACTUAL_CLAIM	Indicates the interest rate associated with the subject, including below a specified threshold.
890	HAS_RISK_TIER	super prime	Yes	FACTUAL_CLAIM	Specifies the risk tier classification related to the subject.
890	HAS_DOWN_PAYMENT	Below 2.0%	Yes	FACTUAL_CLAIM	Indicates the down payment percentage associated with the subject, including below a specified threshold.
335	HAS_LOAN_TERM	12 months	Yes	FACTUAL_CLAIM	Specifies the duration term of the loan associated with the subject.

Canonicalized Triples

Triples after predicate normalization against the database schema. Shows the original predicate, the matched canonical predicate, and match confidence.

Subject	Original Pred.	Object	Canonical Pred.	Confidence
575	HAS_INTEREST_RATE	12.21%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
580	HAS_INTEREST_RATE	11.91%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_INTEREST_RATE	12.03%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
579	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
890	HAS_INTEREST_RATE	Below 4.36%	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	MEDIUM
890	HAS_RISK_TIER	super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
890	HAS_DOWN_PAYMENT	Below 2.0%	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	MEDIUM
335	HAS_LOAN_TERM	12 months	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	HIGH

Logical Inconsistency Report

Contradictions detected between triples in the LLM response — independent of the database. These are self-contradictions within the answer itself.

No logical inconsistencies detected.

Symbolic Rule Violations

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

Constraint ID	Rule Type	Severity	Subject	Predicate	Object	Explanation
c_interest_rate_range		MEDIUM	890	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	Below 4.36 % (qualifier: below threshold)	Subject credit score 890 is outside allowed range 300 to 850.
c_tier_300_579		MEDIUM	890	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	super prime	Subject credit score 890 is outside allowed range 300 to 579.

Detected Hallucinations

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation / Evidence
575	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT	SEMAN TIC		■■■■■ ■■■ 1.00	Object '12.21%' matches DB candidate '12.21' as a semantic equivalent percentage. Evidence: subject: 575, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 12.21
580	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT	SEMAN TIC		■■■■■ ■■■ 1.00	Object '11.91%' matches DB candidate '11.91' as a semantic equivalent percentage. Evidence: subject: 580, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 11.91
579	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.03%	FABRICATED_FACT	NONE	SUBJECT	■■■■■ ■■■ 0.00	No DB candidates found for subject '579' with this predicate indicating fabricated fact.
579	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	PREDICATE_MISMATCH	STRICT	PREDICATE	■■■■■ ■■■ 0.60	Predicate exists for other subjects with object 'deep subprime' but no such predicate for subject '579'. Evidence: subject: 375, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 380, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime
890	DOWN_PAYMENT_REQUIRED_A_S_UPFRONT_BORROWER_CASH_CONTRIBUTION	Below 2.0% (qualifier: below threshold)	FABRICATED_FACT	NONE	SUBJECT	■■■■■ ■■■ 0.00	No DB candidates found for subject '890' with this predicate indicating fabricated fact.
335	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	12 months	PREDICATE_MISMATCH	STRICT	PREDICATE	■■■■■ ■■■ 0.60	Subject '335' exists but predicate 'LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED' not found for it. Evidence: subject: 335, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime; subject: 335, predicate: INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE, object: 20.88; subject: 335, predicate: SCORE_LADDER_ADJACENT_TO_NEXT_HIGHER_FIVE_POINT_BAND, object: 340

Triple Audit Trail (Original → Canonical → Verdict)

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict (e.g. dropped due to output truncation).

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
575	HAS_INTEREST_RATE	12.21%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.21%	CORRECT
580	HAS_INTEREST_RATE	11.91%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	11.91%	CORRECT
579	HAS_INTEREST_RATE	12.03%	HIGH	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	12.03%	FABRICATED_FACT
579	HAS_RISK_TIER	deep subprime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	PREDICATE_MISMATCH
890	HAS_DOWN_PAYMENT	Below 2.0%	MEDIUM	DOWN_PAYMENT_REQUIRED_AS_UPFRONT_BORROWER_CASH_CONTRIBUTION	Below 2.0% (qualifier: below threshold)	FABRICATED_FACT
335	HAS_LOAN_TERM	12 months	HIGH	LOAN_TERM_GRANTED_AS_MAXIMUM_REPAYMENT_WINDOW_OFFERED	12 months	PREDICATE_MISMATCH

Verdict Legend

Verdict	Description
CORRECT	All three components (subject, predicate, object) verified against DB.
RULE_VIOLATION	One or more components contradict DB evidence.
MULTI_RULE_VIOLATION	Violation spans multiple DB triples simultaneously.
UNVERIFIABLE	No usable DB evidence found for any component.
PREDICATE_MISMATCH	Subject real, but stated predicate is not a valid relation for it.
FABRICATED_FACT	Triple asserts a relationship that does not exist anywhere in DB.